



March 11, 2026

District Department of Transportation
250 M Street SE
Washington, DC 20003

Dear DDOT,

An estimated 7 million Americans have low vision—visual acuity typically between 20/70 and 20/200—making it functionally impossible to read a small sign mounted at pole height in the real-world conditions of glare, visual clutter, and distance that define a city sidewalk. This is the “Last Ten Feet” problem: GPS and transit apps get a rider to the block, but the last step—identifying the correct stop and route from a 3-inch character on a pole 8 feet up—fails.

As Chair of the Multimodal Accessibility Advisory Council, established under D.C. Official Code § 50-2361.31, I am writing to propose that DDOT pilot a Horizontal Wayfinding System—high-visibility pavement markings at bus stops—at three to five locations in the District. The concept is simple: put the information where low-vision pedestrians are already looking.

The Science of Where People Look

Eye-tracking research on mobility-impaired pedestrians reveals a consistent pattern. People with low vision do not scan the horizon, where signs are mounted. They scan the ground plane, 5 to 15 feet ahead, to detect tripping hazards, grade changes, and obstacles. This is not a preference. It is a survival behavior trained by years of navigating a built environment that offers no forgiveness for a missed curb or an unseen pothole.

Horizontal wayfinding exploits this existing behavior. A high-contrast marker on the sidewalk—visible in the safety scan zone—delivers boarding information without requiring the traveler to stop, crane upward, and search for a pole. The cognitive load drops. The information enters the visual field incidentally, the same way a curb cut or a tactile warning strip does.

The optics reinforce this. A 3-inch character on a pole 10 feet away subtends a tiny visual angle and demands high acuity to resolve. A 36-inch character on the ground, viewed from standing height at 6 to 10 feet, creates a retinal image large enough that even significant blur cannot destroy the letterform. A “D” is still recognizable as a “D” when it is three feet tall. It is not when it is three inches tall and backlit by a cloudy sky.

Contrast matters as much as size. A dirty white metal sign against an overcast sky produces a contrast ratio of roughly 1.5:1. Dark thermoplastic on light concrete exceeds 3:1—the ISO 23599 standard for visibility. And unlike vertical signs, which reflect sunlight and wash out, matte pavement markings are largely immune to specular glare.

Other Cities Have Done This

Barcelona redesigned its entire bus network in 2012 around an orthogonal grid—Diagonal, Horizontal, and Vertical routes, each color-coded. At interchange hubs where these routes intersect, Transports Metropolitans de Barcelona embedded large colored circles with route numbers and directional arrows directly into the pavement. These are not decorations. They are “transfer assist” devices: a passenger stepping off an H-route looks down, finds the green circle with the V-number, and follows the arrow to the connecting stop around the corner. The system works because it puts information where the decision happens—at foot level, in the path of travel.¹



Barcelona’s Nova Xarxa horizontal wayfinding in practice: route V13 (to Pla. de Palau) and V15 pavement circles at a transfer point. The green circles with route numbers and destinations are visible from standing height—exactly the ground-level information delivery this letter proposes for DC Metrobus stops.

Hong Kong uses sidewalk markings for queue management at high-ridership stops, demonstrating that thermoplastic survives millions of annual footfalls when maintained. Their “LOOK LEFT” / “LOOK RIGHT” crosswalk text establishes a broader precedent: safety-critical information on the ground is effective at managing pedestrian attention.²

San Francisco’s SFMTA places a square universal symbol of accessibility directly on the sidewalk at constrained bus stops so that passengers know exactly where the boarding ramp will deploy—

¹TMB, “How to travel in the new bus network,” <https://www.tmb.cat/en/get-to-know-tmb/transport-network-improvements/new-bus-network/how-to-use-it>. The interchange area guide shows colored route circles with directional arrows embedded in the pavement to guide transfers between bus lines.

²Hong Kong Transport Department, “Using Buses,” https://www.td.gov.hk/en/road_safety/road_users_code/index/chapter_3_for_passengers/using_buses_light_rail_vehicles/index.html.

a ground-level wayfinding element already in use in a US city.³

Closer to DC’s regulatory environment, Seattle’s “Seamless Seattle” pilot used inlaid medallions and tactile indicators—hardware, not paint—to mark accessible routes and confirm boarding locations. The preference for durable, engineered materials over temporary coatings aligns with the Public Space Committee’s emphasis on quality and permanence in the District’s public realm.⁴

The Regulatory Path: Traffic Control, Not Art

DDOT’s Art in the Right of Way (AROW) program explicitly prohibits text of any kind in public art installations. This is a content-neutrality safeguard rooted in First Amendment law—if DDOT permits “D4” as art, it cannot deny a political slogan or an advertisement under the same permit class. The prohibition is correct, and it means the AROW pathway is closed.

The viable path is to classify horizontal wayfinding markings as Traffic Control Devices. The MUTCD 11th Edition (December 2023), Section 3B.20, allows pavement text for regulatory, warning, and guidance purposes, but there is no existing standard for transit route markings on sidewalks. This means DDOT would need to submit a Request to Experiment to the Federal Highway Administration under MUTCD 11th Edition, Section 1B.05.

An FHWA experimentation request must include a problem statement, device description with illustrations, supporting research data, a comparison to existing compliant alternatives, a project timeline and location, control sites, an evaluation methodology, a commitment to semi-annual progress reports, and a site restoration agreement. A 3-to-5-stop pilot is the right scale for this. It is small enough to manage administratively, large enough to generate statistically meaningful data, and precisely the kind of evidence-based initiative that FHWA approves routinely.

Why Paint Fails and What Works Instead

The material question is not aesthetic. It is a matter of slip-and-fall liability.

Material	BPN (Wet)	Durability	Visual Definition
Standard Traffic Paint	40–50	<1 year	Fuzzy, fading
Methyl Methacrylate (MMA)	60–70	5–8 years	Sharp
Preformed Thermoplastic	60–70	3–6 years	Very sharp (factory cut)
Safety Threshold (BPN)	≥65		

Standard traffic paint fills the micro-texture of concrete, creating a hydroplaning surface when wet. Its British Pendulum Number drops to 40–50—well below the 65 BPN safety threshold that AASHTO and industry practice recommend for pedestrian zones. Glass beads added for

³SFMTA, “Accessible Signage and Wayfinding,” <https://www.sfmta.com/accessibility-strategy-needs-assessment-2024/muni-capital-projects/34-accessible-signage-and-wayfinding>.

⁴NACTO, “Passenger Information & Wayfinding,” <https://nacto.org/publication/transit-street-design-guide/station-stop-elements/stop-elements/passenger-information-wayfinding/>.

retroreflectivity make it worse for foot traffic. And it wears off in six months, leaving a “ghost” that looks neglected.

Preformed thermoplastic with embedded corundum or calcined bauxite aggregate is the appropriate material. It fuses mechanically with the concrete surface. As the top layer wears, new anti-skid aggregate is exposed, maintaining friction throughout the product’s 3-to-6-year life. Factory-cut letterforms produce crisp, engineered edges—not the hand-painted look that triggers visual clutter objections.

For ADA compliance, the markings must be flush with the surrounding concrete to avoid creating a false tactile signal. A cane user must not mistake a raised thermoplastic edge for a truncated dome warning field, which signals a hazardous street edge. Inlaying the material into a shallow recess eliminates this risk.

What We Are Proposing

A small, controlled pilot at three to five bus stops, designed to generate the data needed for a Request to Experiment and to answer two questions: Does horizontal wayfinding reduce “Time to Board” for low-vision riders? And does it improve navigational confidence compared to the existing vertical-only signage?

The critical design element is this: the markings must display the route number and direction of travel. A generic “Bus Stop” marker solves none of the problems described in this letter. The entire point is that a low-vision rider standing on a sidewalk cannot read the route number on the pole. A ground marking that also omits the route number is just a larger version of the same failure. The rider needs to know *which* bus stops here and *where* it is going—from the ground, at a glance, without stopping to search.

The proposed design is a high-contrast circle, approximately 3 feet in diameter, displaying the Metrobus route number and terminal destination, using WMATA’s existing brand colors. Each side of the street receives only the direction served by that stop. A mockup of the proposed marking is shown below.



Proposed pavement marking: route number and destination in a high-contrast circle using WMATA brand colors (Metrobus Red #BF0D3E). Approximately 3 feet in diameter, applied in preformed thermoplastic at the boarding zone. Each stop shows only the direction it serves.

At stops serving multiple routes, multiple circles would be installed in the boarding zone—one per route, each displaying its route number and terminal destination. This follows the Barcelona model, where interchange hubs use one colored circle per route to guide transfers.

The “ghosting” concern—that route-specific markings become wrong when service changes—is real but manageable. The Better Bus network redesign is happening now, and pilot stops should be selected on corridors where the new route designations are already finalized. Thermoplastic has a 3-to-6-year service life; when a route changes, the marking is due for replacement regardless. The alternative—a generic marker that says nothing about which route stops here—defeats the purpose of the intervention entirely.

Site selection criteria:

1. **Near a medical facility or rehabilitation center**—where the proportion of riders with mobility and vision impairments is highest. A stop serving Howard University Hospital, MedStar Washington Hospital Center, or the Department of Veterans Affairs Medical Center would be ideal.
2. **A major transfer hub**—where multiple routes converge and the wayfinding challenge is most acute. Gallery Place, Anacostia Metro station, or Columbia Heights are candidates.
3. **A Better Bus redesign corridor**—to align the pilot with DDOT’s existing capital program and avoid duplicating infrastructure investment.
4. **A residential stop in a high-disability-population ward**—to test the intervention outside the downtown core, where ridership patterns, sidewalk conditions, and lighting differ.
5. **A control stop**—a comparable location that receives no intervention, to isolate the effect of the marking from other variables.

Cost estimate: Preformed thermoplastic for a single 3-foot-diameter circle costs approximately \$150–\$250 in material, plus installation labor. Site preparation (recessing the concrete for inlay) adds cost, but even with engineering, permitting, survey design, and evaluation, a 5-stop pilot should fall well under \$50,000.

On behalf of the Multimodal Accessibility Advisory Council, we formally resolve:

That DDOT initiate a Horizontal Wayfinding Pilot at three to five bus stops in the District, using high-friction preformed thermoplastic to install route-specific pavement markings—displaying the Metrobus route number and terminal destination in WMATA brand colors—at the boarding zone, and that DDOT submit a Request to Experiment to the Federal Highway Administration to establish the regulatory basis for broader deployment.

We request the following specific actions:

1. Submit a Request to Experiment to the FHWA under MUTCD 11th Edition, Section 1B.05, for “Sidewalk Wayfinding Markings for Transit Boarding Zones,” incorporating a problem statement, device description, supporting data, evaluation methodology, control sites, and site restoration commitment, within 90 days.
2. Select 3–5 pilot bus stops using the criteria above—medical facility proximity, transfer hub complexity, Better Bus corridor alignment, and demographic representation—and brief MAAC on the selections within 60 days.
3. Specify preformed thermoplastic with high-friction aggregate ($BPN \geq 65$) as the mandatory material, inlaid flush with the sidewalk surface to prevent false tactile signaling for cane users.
4. Design an evaluation methodology, including “Time to Board” measurement and navigational confidence surveys for low-vision participants, in consultation with the disability community.
5. Require that any utility excavation within a pilot zone restore the marking as a condition of the street-cut permit.
6. Report results to MAAC, the Council’s Committee on Transportation and the Environment, and the FHWA within 12 months of installation, with a recommendation on whether to proceed to broader deployment.

This is not a demand to paint every sidewalk in the city tomorrow. It is a request for a disciplined, small-scale experiment grounded in human factors research, material science, and federal regulatory procedure. Five bus stops. One FHWA filing. The cost is negligible. The potential benefit—a replicable model for solving the Last Ten Feet problem for millions of Americans with low vision—is not.

The bus stops are already there. The thermoplastic exists. The science says it works. Let us find out.

We look forward to your response.

Sincerely,
Dr. Karthik Balasubramanian
Chair, Multimodal Accessibility Advisory Council

CC:

- Federal Highway Administration, Office of Transportation Operations